



```
public class MainActivity extends AppCompatActivity implements View.OnClickListener
```

```
    findViewById(R.id.btnwriteIS).setOnClickListener(this);  
    findViewById(R.id.btnreadIS).setOnClickListener(this);  
    findViewById(R.id.btnwriteFC).setOnClickListener(this);  
    findViewById(R.id.btnreadFC).setOnClickListener(this);  
    findViewById(R.id.btnwriteES).setOnClickListener(this);  
    findViewById(R.id.btnreadES).setOnClickListener(this);
```

@Override

```
public void onClick(View view) {  
    switch (view.getId()){  
        case R.id.btnreadIS:  
            String readDataIS = readIS("myfile.txt");  
            Toast.makeText(this, "Nội dung file:" + readDataIS, Toast.LENGTH_SHORT).show();  
            break;  
        case R.id.btnwriteIS:  
            writeIS("myfile.txt", "Xin chào các bạn!");  
            break;  
        case R.id.btnreadFC:  
            String readDataFC = readFC("mycached.cache");  
            Toast.makeText(this, "Data cache: " + readDataFC, Toast.LENGTH_SHORT).show();  
            break;  
        case R.id.btnwriteFC:  
            writeFC("mycached.cache", "Dữ liệu lưu vào cahce");  
            break;  
        case R.id.btnwriteES:  
            writeES("mysdcard.txt", "Dữ liệu lưu vào thẻ nhớ ngoài");  
            break;  
        case R.id.btnreadES:  
            String readDataEX = readES("mysdcard.txt");  
            Toast.makeText(this, "Data sdcard: " + readDataEX, Toast.LENGTH_SHORT).show();  
            break;  
    }  
}
```

```
private String readIS(String filename) {
    try (FileInputStream fin = openFileInput(filename))
    {
        byte[] buffer = new byte[1024];
        int length = 0;
        length = fin.read(buffer);
        return new String(buffer,0,length);
    } catch ( Exception ex ) {
        ex.printStackTrace();
    }
    return null;
}
```

READ INTERNAL STORAGE

WRITE INTERNAL STORAGE

```
private void writeIS(String filename,String data) {

    try {

        FileOutputStream fout = openFileOutput(filename,MODE_PRIVATE);
        fout.write(data.getBytes(StandardCharsets.UTF_8));
        fout.close();
        Toast.makeText(this, "Dữ liệu đã được ghi",
Toast.LENGTH_SHORT).show();
    } catch ( Exception ex )
    {
        Toast.makeText(this, "Error:" + ex.getMessage(),
Toast.LENGTH_SHORT).show();
        ex.printStackTrace();
    }
}
```

```
private void writeFC(String filename, String data) {
    File cacheDir = getCacheDir();
    try (FileOutputStream fout = new FileOutputStream(new
File(cacheDir,filename))) {
        fout.write(data.getBytes(StandardCharsets.UTF_8));
        Toast.makeText(this, "Dữ liệu lưu vào cache thành
công!", Toast.LENGTH_SHORT).show();
    } catch (Exception ex)
    {
        ex.printStackTrace();
        Toast.makeText(this, "Error:" + ex.getMessage(),
Toast.LENGTH_SHORT).show();
    }
}
```

READ CACHE FILE

WRITE CACHE FILE

```
private String readFC(String filename) {
    File cacheDir = getCacheDir();
    byte[] buffer = new byte[1024];
    int length = 0;
    try (FileInputStream fin= new FileInputStream(new File(cacheDir,
filename))) {
        length = fin.read(buffer);
        return new String(buffer,0,length);
    } catch (Exception ex) {
        ex.printStackTrace();
        Toast.makeText(this, "Error: " + ex.getMessage(),
Toast.LENGTH_SHORT).show();
    }
    return null;
}
```

```
private void writeES(String filename, String data) {

    String sdcard = getExternalFilesDir("") + "/" + filename;

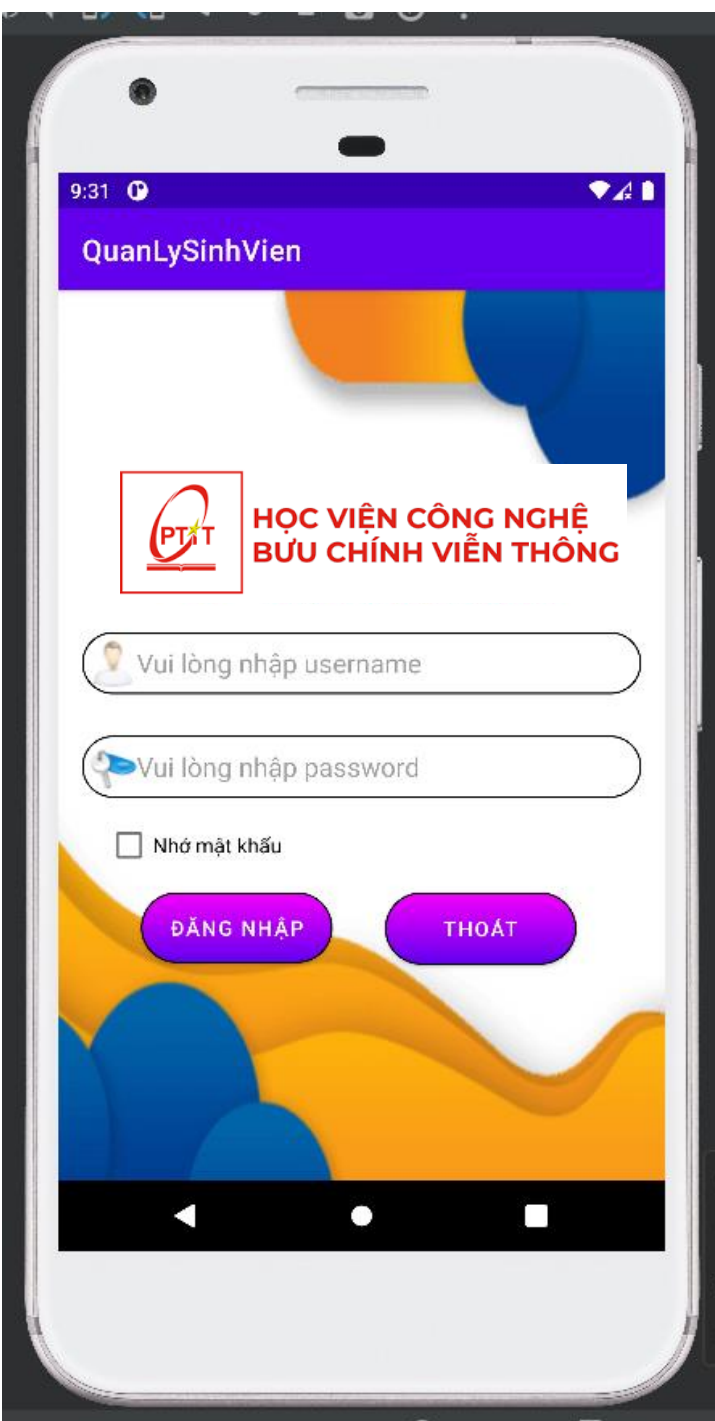
    try {
        OutputStreamWriter writer = new OutputStreamWriter(new
FileOutputStream(sdcard));
        writer.write(data);
        writer.close();
        Toast.makeText(this, "Lưu thành công", Toast.LENGTH_SHORT).show();
    } catch ( Exception ex )
    {
        ex.printStackTrace();
    }
}
```

WRITE EXTERNAL STORAGE

READ EXTERNAL STORAGE

```
private String readES(String filename) {

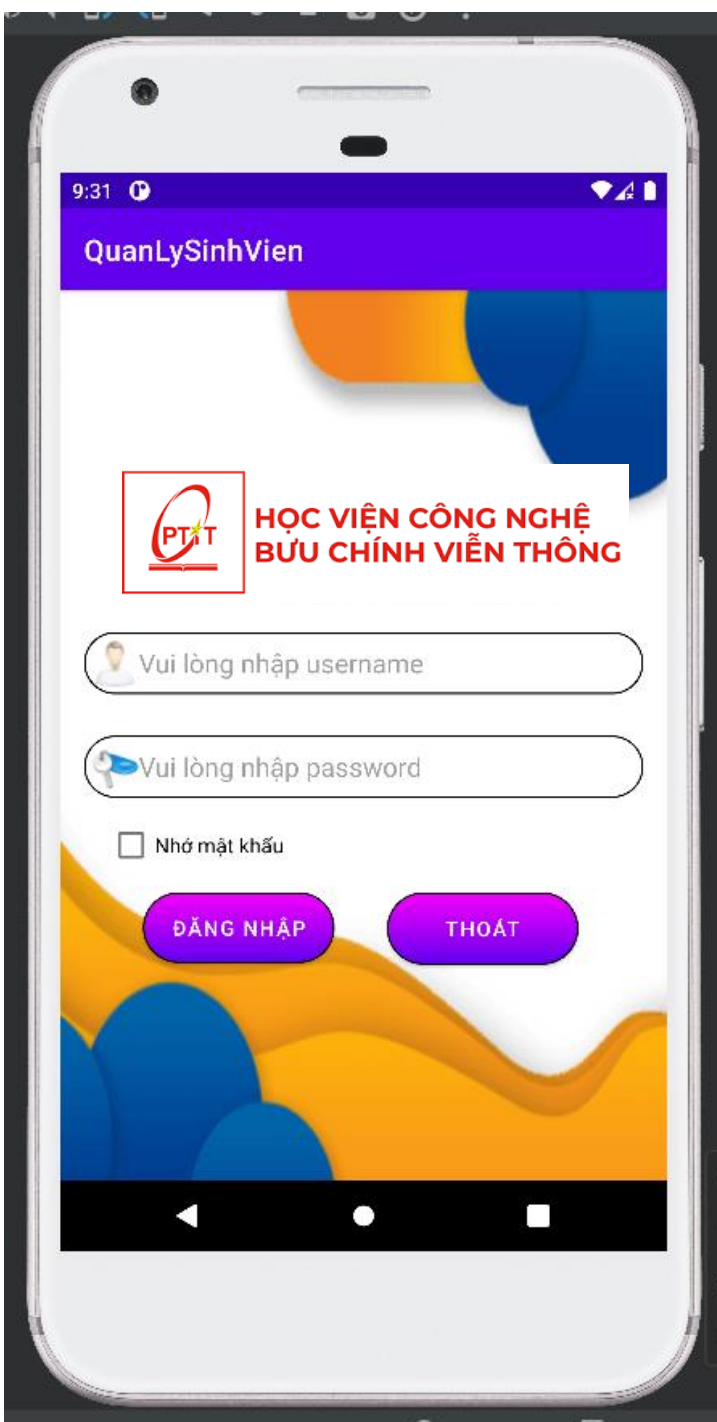
    String sdcard = getExternalFilesDir("") + "/" + filename;
    try {
        Scanner scan = new Scanner(new File(sdcard));
        String data="";
        while (scan.hasNext()){
            data = data + scan.nextLine() + "\n";
        }
        scan.close();
        return data;
    } catch ( Exception ex )
    {
        ex.printStackTrace();
    }
}
```



```
public class LoginActivity extends AppCompatActivity implements View.OnClickListener
```

```
    findViewById(R.id.btnLogin).setOnClickListener(this);  
    findViewById(R.id.btnThoat).setOnClickListener(this);  
    edtUsername = findViewById(R.id.edtUsername);  
    edtPassword = findViewById(R.id.edtPassword);  
    ckRememberPassword = findViewById(R.id.ckRememberPassword);
```

```
    private void saveAutoLogin(){  
        SharedPreferences preferences = getSharedPreferences("Login", MODE_PRIVATE);  
        SharedPreferences.Editor editor = preferences.edit();  
        editor.putString("username",edtUsername.getText().toString());  
        editor.putString("password",edtPassword.getText().toString());  
        editor.commit();  
    }
```



```
private boolean readAutoLogin(){
    SharedPreferences preferences = getSharedPreferences("Login", MODE_PRIVATE);
    String username = preferences.getString("username", "");

    if(username != null && !username.equals(""))
    {
        Intent intent = new Intent(this, MainActivity.class);
        startActivity(intent);
        finish();
        return true;
    }
    return false;
}
```

```
private void clearAutoLogin() {
    SharedPreferences preferences = getSharedPreferences("Login", MODE_PRIVATE);
    SharedPreferences.Editor editor = preferences.edit();
    editor.clear();
    editor.commit();
}
```